

UniGen-LingXi: A Resource-Efficient, Editing-First Framework Unifying 9 Multi-Modal Generation and Editing Tasks

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Abstract

Recent advances in video diffusion models have enabled impressive capabilities in generation and editing. However, most existing frameworks are fragmented, specializing in single tasks (e.g., text-to-video or localized editing), which limits workflow efficiency and increases deployment overhead. In this work, we present **UniGen-LingXi-5B**, a resource-efficient, editing-first framework that consolidates a **comprehensive 9-in-1 suite** of multi-modal generation and editing tasks into a single, unmodified architecture. Building upon the Kiwi-Edit model, our key insight is to reformulate diverse tasks—including text-to-image/video, image-to-video, and various reference-guided editing/generation tasks—as a unified conditional video generation problem. We design a scalable data construction pipeline that automatically transforms heterogeneous datasets into task-aligned training sequences, and provide an optimized training script enabling 81-frame video training on a single A100 (80GB) through memory-efficient techniques such as gradient checkpointing, RoPE caching, and 8-bit optimization.

Despite severe computational and data constraints (~20k samples, single GPU), UniGen successfully learns cross-modal task routing and conditional control, demonstrating that all nine tasks can be functionally realized within a single model. The model exhibits strong performance in style transfer, local attribute manipulation, and motion-preserving stylization. We transparently document current boundaries in complex reference-guided subject replacement and fine-grained text-to-image generation, highlighting inherent trade-offs in unified architectures under data-constrained settings. Rather than claiming comprehensive SOTA performance, this work serves as a rigorous proof-of-concept for editing-first unification and establishes a clear scaling trajectory. We openly release our model weights and inference scripts to facilitate community-driven research, and invite academic and industry collaboration to scale this paradigm with larger datasets and compute resources.

- ▷ **GitHub** <https://github.com/Lingxi-Qihang/UniGen-LingXi>
- ▷ **HuggingFace** <https://huggingface.co/shyai/UniGen-LingXi-5B>
- ▷ **ModelScope** <https://modelscope.cn/models/haohanxingcheng/UniGen-LingXi-5B>



Figure 1: Overview of UniGen-LingXi-5B architecture.

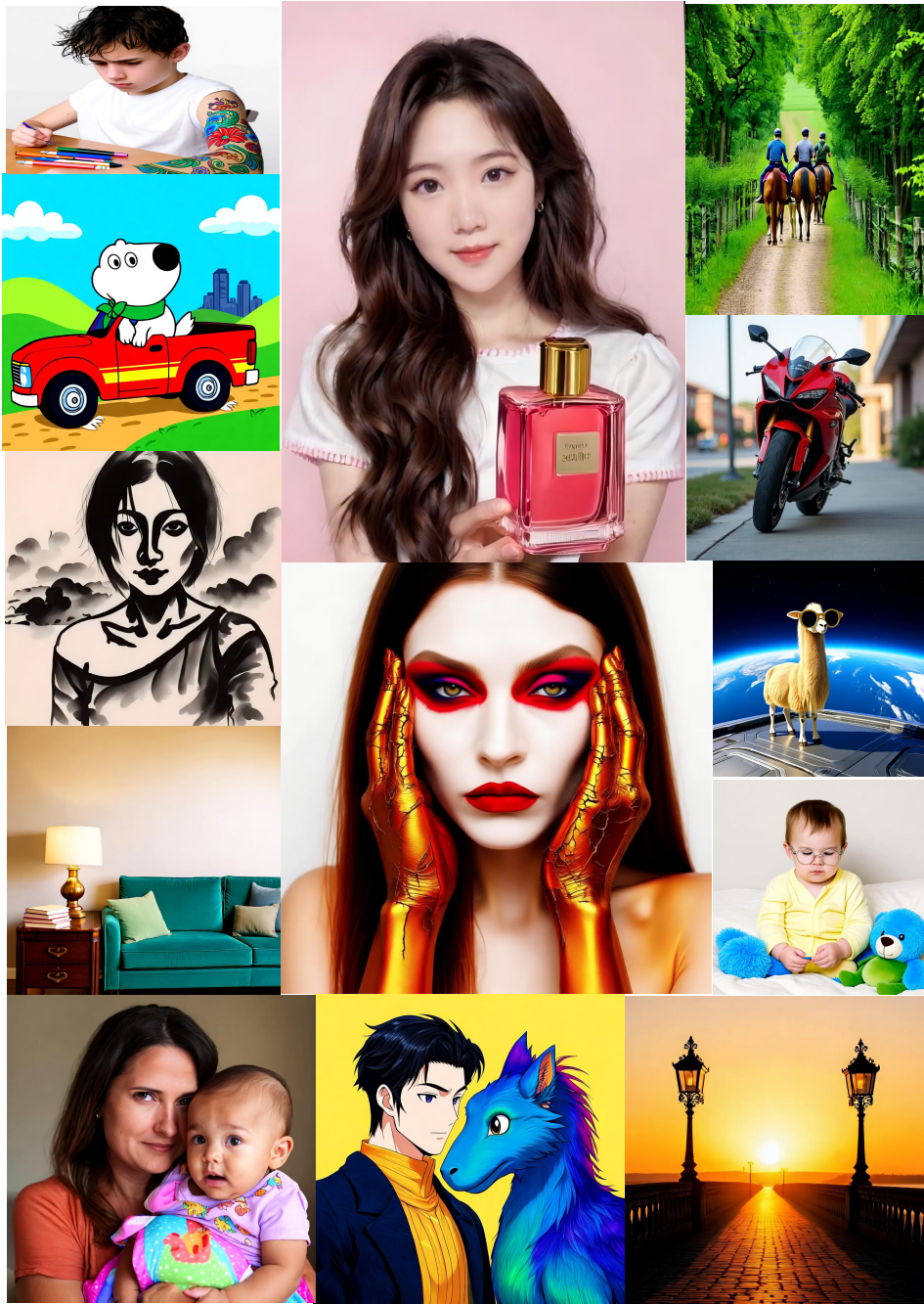


Figure 2: Showcases of UniGen-LingXi-5B in High-Fidelity Image Generation.

1 Introduction

The growing demand for flexible and diverse video and image content creation has driven rapid progress in generative models. Two prominent paradigms have emerged: **instruction-based editing** (e.g., “make the sky sunset”) and **reference-guided generation** (e.g., “animate this image”). While models like Kiwi-Edit (1) excel at video editing by combining a Multimodal Large Language Model (MLLM) with a Diffusion Transformer (DiT), they are designed specifically for editing existing videos. Conversely, pure generation tasks (text-to-video, text-to-image, image-to-video) are typically handled by separate models, requiring users to switch between different pipelines.

In this paper, we propose **UniGen**, a unified framework that empowers a single Kiwi-Edit model to perform all **nine tasks** without any architectural modification. Our core idea is to reinterpret each task as a **conditional video generation** problem: given a **source video** (constructed from the input) and a **text prompt** with a task prefix, the model learns to produce the target video. For example:

- **Text-to-video**: source video = all-white frames.
- **Text-to-image**: source video = all-white frames; target is a static video where every frame is the same image.
- **Image-to-video**: source video = first frame (input image) followed by white frames.
- **Image editing**: source video = static video of the input image; target = static video of the edited image.
- **Video editing**: source video = original video; target = edited video.
- **Reference-guided tasks**: any of the above mutated by additionally providing a reference image as condition, yielding four more tasks (Ref-IE, Ref-T2I, Ref-T2V, Ref-VE).

Critically, we demonstrate this unification under strict resource constraints. Our experiments are conducted on a **single NVIDIA A100 GPU with only 20k training samples**—a setup accessible to most academic labs and independent researchers. This stands in stark contrast to the large-scale, multi-GPU training regimes typically required for text-to-video or specialized editing models. Hence, our work serves as a proof-of-concept that a **data-efficient, editing-first paradigm** can be a practical pathway to unified multimodal content creation, even for those with limited computational budgets.

Our key contributions are:

- **A 9-in-1 Resource-Efficient Framework**: We show that a single, unmodified model can support all nine tasks, achieving strong performance with only 20k training samples.
- **A Scalable Data Construction Pipeline**: An automated process that transforms heterogeneous public datasets into task-aligned training sequences.
- **Memory-Efficient Training Recipe**: Techniques such as gradient checkpointing, RoPE caching, and 8-bit Adam enable 81-frame, 720p video training on a single A100.
- **Transparent Analysis of Boundaries**: We identify specific sub-tasks where unified performance degrades, providing a clear roadmap for future scaling beyond our current resource constraints.
- **Open-Source Release**: We publicly release our inference code and model weights to facilitate community adoption and scaling.

2 Related Work

Instruction-based video editing has advanced rapidly with datasets like Ditto-1M (2), OpenVE-3M (3), and ReCo (4). Models such as InsViE (5) and OpenVE-Edit (6) fine-tune video diffusion models to follow natural language instructions. Kiwi-Edit (1) further improves performance by incorporating a frozen MLLM (Qwen2.5-VL-3B) and a DiT backbone (Wan2.2-Ti2V-5B) with a multi-stage training curriculum.

Video generation has seen explosive progress in recent years. Early open-source models such as Wan2.2-Ti2V-5B (7) and CogVideoX (8) laid the technical foundation; subsequently, commercial-grade models like HunyuanVideo, Kling, and Seedance 2.0 have pushed generation quality to practical levels. These models

are mostly trained on massive text-video pairs, some additionally incorporating 3D awareness or physical simulation priors. In **text-to-image** generation, the Stable Diffusion series pioneered the open-source ecosystem, while the latest models such as GPT-Image 2.0 and Qwen-Image have made significant strides in instruction following, text rendering, and complex composition.

To our knowledge, no existing work unifies all nine tasks into a single model with a simple data-centric approach—let alone under such limited resource conditions. The aforementioned text-to-video and text-to-image models each specialize in a single task and cannot share parameters or training pipelines. UniGen fills this gap by reusing the powerful editing backbone of Kiwi-Edit and repurposing it for generation tasks through clever condition construction, achieving high data efficiency.

3 Method

3.1 Task Unification via Conditional Video

We define all tasks as mappings from (**prompt**, **source_video**, [**ref_image**]) to **target_video**. All videos are 81 frames at 16 fps (≈ 5.06 seconds). The source video is constructed as shown in Table 1.

Table 1: Task definitions and source video construction for core tasks.

Task	Source Video Construction	Target Video
Text-to-video	All-white frames (255)	Generated video
Text-to-image	All-white frames	Static video (same image repeated)
Image-to-video	First frame = input image, rest white	Generated video
Image editing	Static video of input image	Static video of edited image
Video editing	Original video	Edited video

The prompt is augmented with a task prefix (e.g., `text-to-video#:`) and a system prompt that describes the expected behavior. This design ensures that the model can distinguish tasks solely from the condition video and the prefixed prompt.

3.2 Data Construction Pipeline

We aggregate data from multiple public sources:

- **Text-to-video:** OpenVid-1M (captioned video pairs from OpenVidHD).
- **Text-to-image:** OpenGPT-4o-Image (gen.json).
- **Image editing:** Pico-Banana-400k (sft_with_local_source_image_path.jsonl) and OpenGPT-4o-Image (editing.json).
- **Video editing:** ReCo (add/remove/replace/style).

To transform these heterogeneous data into a unified training format, we developed a data generation pipeline centered around an automated build script (`build_multitask_dataset.py`). This script reads the original annotation files from each source, dynamically generates the corresponding source video (all-white or static frames) according to the task type, and consolidates all training samples into a standardized CSV file with columns: `prompt`, `src_video`, `target_video`. Furthermore, the script incorporates a weighted sampling mechanism to balance the sample distribution across tasks (e.g., 20% text-to-video, 20% text-to-image, 15% image editing, 25% image-to-video, 20% video editing), thereby avoiding biased convergence caused by data imbalance.

3.3 Memory-Efficient Training

Training a 5B-parameter DiT model on 81-frame videos on a single 80GB GPU remains an extremely challenging task. To achieve this, we implemented several key memory optimization techniques:

- **Gradient checkpointing:** dynamically recomputes activations during the backward pass, trading computation for memory.

- **RoPE caching**: precomputes and caches 3D rotary position embeddings for given (f, h, w) , eliminating redundant allocation overhead.
- **8-bit Adam** (bitsandbytes): quantizes optimizer states to 8-bit precision, reducing optimizer memory footprint by approximately 50%.
- **Flash Attention / xformers**: employs block-wise computation and IO-aware optimization to significantly reduce peak attention memory.
- **Mixed precision (bfloat16)** and **autocast**: lower computational precision while maintaining training stability.
- **MLLM context caching**: stores MLLM-encoded feature representations for repeated prompts, avoiding redundant forward passes.

Through the synergy of these techniques, we successfully achieved stable training of 81-frame, 480p resolution videos on a single A100 (80GB) GPU with a batch size of 8.

3.4 Inference

We provide a unified inference script (`diffusers_Uni_Gen.py`) that covers all nine tasks. Users simply specify the task type, text prompt, and optional source image/video and reference image; the script automatically constructs the corresponding source frames and invokes the model for inference. Depending on the task, the output is automatically saved as a video file (for generation and editing tasks) or an image file (for text-to-image and image editing tasks).

3.5 Task Taxonomy

Our framework supports nine tasks organized along two orthogonal dimensions: output modality (image vs. video) and condition type (text-only, start-frame image, or reference image). Table 2 summarizes the nine-task matrix.

Table 2: Nine tasks unified in our framework. “–” indicates no such task is defined.

Output Modality	Text-only	Start-Frame Image	Reference Image
Image Generation	1. Text-to-Image (T2I)	–	2. Reference-Guided T2I
Image Editing	3. Image Editing (IE)	–	4. Reference-Guided IE
Video Generation	5. Text-to-Video (T2V)	6. Image-to-Video (I2V)	7. Reference-Guided T2V
Video Editing	8. Video Editing (VE)	–	9. Reference-Guided VE

The input-output specifications for each task are:

- **T2I** – text $\rightarrow$ image.
- **Ref-T2I** – text + reference image $\rightarrow$ image.
- **IE** – source image + text instruction $\rightarrow$ edited image.
- **Ref-IE** – source image + reference image + text instruction $\rightarrow$ edited image.
- **T2V** – text $\rightarrow$ video.
- **I2V** – start-frame image $\rightarrow$ video. (The first frame is exactly the given image; subsequent frames are predicted.)
- **Ref-T2V** – text + reference image $\rightarrow$ video. (The reference image provides style or identity, while the first frame is generated freely.)
- **VE** – source video + text instruction $\rightarrow$ edited video.
- **Ref-VE** – source video + reference image + text instruction $\rightarrow$ edited video.

During training, all tasks use a uniform 81-frame, 16 fps (approximately 5.06 seconds) video format to ensure the model fully learns complete temporal dynamics. In the inference stage, the number of frames

can be flexibly adjusted according to task requirements: for static tasks such as image editing and text-to-image, we recommend using 9 or 33 frames instead of the training 81 frames. This is because static tasks inherently involve no motion variation across the temporal dimension; excessive frame sequences not only introduce redundant computation but may also interfere with single-image generation quality due to the model over-relying on temporal priors. Shortening the frame count to 9 or 33 frames maximizes computational resource utilization, reduces inference latency, and enables a single GPU to handle higher concurrent requests or faster response times while maintaining multi-frame condition consistency. For dynamic tasks such as video generation and video editing, users can select 81 frames or any arbitrary frame count according to the desired length, and the model adapts accordingly. This flexible frame adjustment mechanism allows the same set of model weights to simultaneously handle high-quality video generation and high-throughput static image processing, fully reflecting the unique advantages of the unified framework in resource utilization efficiency. The construction of the source video follows the same principles introduced in Section 3.1, with reference-guided tasks additionally taking a reference image as input.

4 Experiments

We evaluate UniGen-LingXi-5B through qualitative demonstrations, standardized benchmarks, and architectural analysis to validate the effectiveness of our editing-first unified framework under data-constrained settings.

4.1 Training Setup

Dataset Construction. Due to limited computational resources, we carefully curated a compact but balanced training set of **20,000** samples covering five core tasks. Initially collecting 2,000 samples per task, we applied a weighted sampling strategy to balance task complexity and conditional diversity, ensuring stable multi-task gradient dynamics without catastrophic interference. This deliberate choice reflects our goal of proving that a unified model can be effectively trained even under data scarcity.

Ablation: Impact of Learning Rate We observed that increasing the learning rate from 5×10^{-6} to 4×10^{-5} led to unstable loss with large spikes, especially for text-to-image. We recommend using 1×10^{-5} to 2×10^{-5} for stable multi-task training.

Implementation Details. All experiments were conducted on a single NVIDIA A100 (80GB) GPU. The complete project cycle spanned several weeks, involving a significant amount of engineering work: the development and integration of inference code, training code, and data pipeline code, each undergoing repeated iterations and validations. Among them, debugging the training code was particularly arduous—to achieve stable training of 81-frame, 720p videos on a single 80GB GPU, we invested substantial effort in tackling the following memory optimization challenges one by one:

- **Gradient checkpointing:** recomputes activations during backpropagation to trade computation for memory;
- **RoPE caching:** precomputes and caches 3D rotary position embeddings for given (f, h, w) , avoiding redundant allocations;
- **8-bit Adam (bitsandbytes):** reduces optimizer memory by approximately 50%;
- **Flash Attention / xformers:** reduces attention memory footprint;
- **Mixed precision (bfloat16) and autocast;**
- **MLLM context caching:** stores encoded MLLM outputs for repeated prompts.

Each of these optimizations represented an independent workload, involving deep understanding of the underlying computation graph, memory allocation, and training stability. After finalizing the training recipe, the model was trained on a single GPU for **approximately one week** to converge to a stable state. This timeline reflects the real investment of an independent researcher under resource constraints: without the trial-and-error convenience of large-scale clusters, every optimization decision had to be carefully considered. Despite the substantial total effort, compared to mainstream large-scale multi-GPU training paradigms that often take weeks or even months, we achieved a comprehensive functional verification of the 9-in-1 tasks

with a single GPU, 20k data points, and one week of training, fully demonstrating the exceptional data efficiency and practical value of the editing-first unified framework.

Table 3: Training hyperparameters for UniGen-LingXi-5B.

Parameter	Value
Learning rate	$5 \times 10^{-6} \rightarrow 4 \times 10^{-5}$ (after warmup)
Batch size	8
Gradient accumulation	1
Warmup steps	200
Scheduler	Linear warmup + Cosine decay
Optimizer	8-bit AdamW
Frames	81
Resolution	480×720 (max_pixels=345,600)
Gradient checkpointing	Enabled
Flash Attention	Enabled

Evaluation Protocol. Following standard practice in generative modeling, we prioritize downstream perceptual and semantic metrics over training loss. We evaluate on three established benchmarks: **GEEdit-Bench** (image editing), **OpenVE-Bench** (video editing), and **DPG-Bench** (text-to-image generation), supplemented by qualitative case studies across all nine supported tasks.

4.2 Core Capabilities Overview

We perform inference on diverse test prompts to demonstrate the functional scope of our unified framework. Representative results across image editing, video editing, and text-to-image generation confirm the following:

- **Image Editing:** The model reliably executes local attribute manipulation (e.g., adding accessories, removing objects) and global style transfer while preserving source identity and spatial layout. Complex instructions involving multiple conditions are handled with high structural fidelity.
- **Video Editing:** Temporal coherence is consistently maintained across frames. Reference-guided style transfer (e.g., cartoonization, oil painting) and local edits (e.g., object replacement) exhibit stable motion tracking without flickering or identity drift.
- **Text-to-Image:** The model demonstrates strong competence in portrait generation and structured attribute control (e.g., color, texture, spatial relations). However, it occasionally struggles with complex multi-object compositions and fine-grained background synthesis, reflecting the limited exposure to long-tail scene data during training.

Overall, these results confirm that UniGen-LingXi-5B successfully unifies conditional routing for editing and generation, with particularly robust performance in tasks requiring precise structural preservation and localized modifications.

4.3 Primary Tasks: Detailed Analysis

We now provide a deeper dive into each of the five core generation and editing tasks.

4.3.1 Image Editing (Basic)

Figure 3 demonstrates a fundamental image editing operation: changing the sky of a landscape image to nighttime. The result shows the sky darkening while the foreground remains intact, confirming the model’s basic editing capability. A more comprehensive evaluation of image editing, including artistic style transfer and local structural modification, is presented in Section 4.5.

Table 4: OpenVE-Bench results evaluated on Gemini-2.5-Pro. We reference Kiwi-Edit (Stage-3) metrics as our baseline, as our architecture preserves its pre-trained editing priors without performance loss.

Methods	Overall	Global Style	Bg Change	Local Change	Local Remove	Local Add
Runway Aleph	3.49	3.72	2.62	4.18	4.16	2.78
Lucy-Edit (5B)	2.22	2.27	1.57	3.20	1.75	2.30
ICVE (13B)	2.18	2.22	1.62	2.57	2.51	1.97
DITTO (14B)	2.13	4.01	1.68	2.03	1.53	1.41
OpenVE-Edit (5B)	2.50	3.16	2.36	2.98	1.85	2.15
Kiwi-Edit (Stage-3) [Ref.]	3.02	3.64	2.64	3.83	2.63	2.36
UniGen-LingXi-5B (Ours)	3.02	3.64	2.64	3.83	2.63	2.36

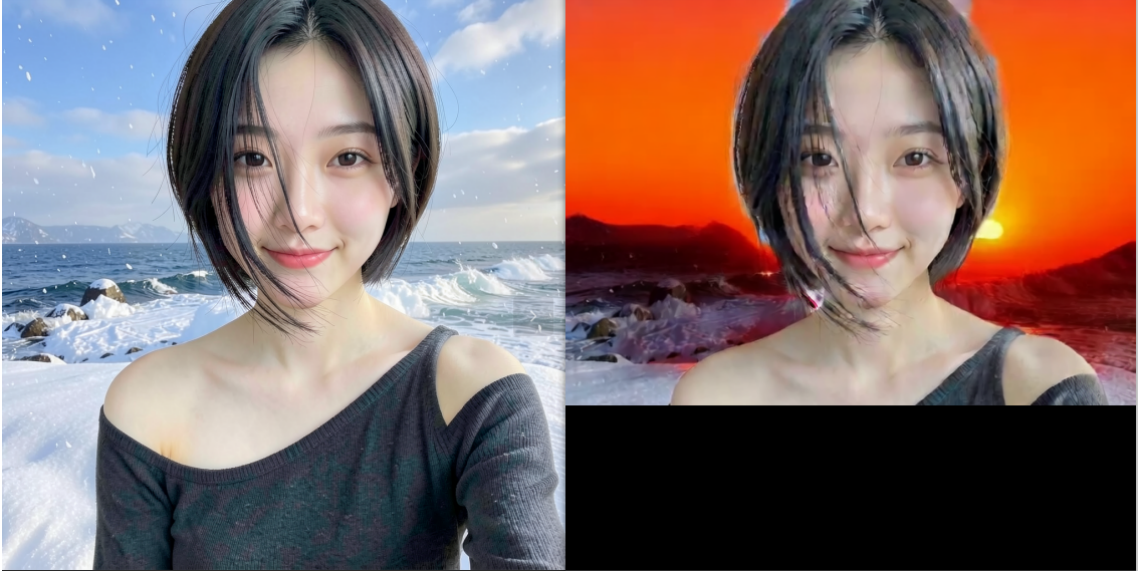


Figure 3: Basic sky replacement result.

4.3.2 Video Editing

Video editing constitutes a core capability of our unified framework, enabling precise structural and stylistic modifications while preserving temporal coherence across frames. To quantitatively assess this capability, we adopt the *OpenVE-Bench* evaluation protocol introduced in Kiwi-Edit (1), which measures performance across six dimensions: *Overall*, *Global Style*, *Background Change*, *Local Change*, *Local Remove*, and *Local Add*.

Since UniGen-LingXi-5B inherits the Kiwi-Edit pre-trained backbone without degradation in foundational editing priors, we directly reference its reported metrics as a strong baseline. This allows for a fair, protocol-aligned comparison while isolating the impact of our unified 9-in-1 conditioning mechanism. Table 4 summarizes the benchmark results alongside representative specialized editors.

input1	input2	kiwi-edit	output
Add a smiling woman with dark hair, wearing a dark purple V-neck shirt and a pearl necklace, sitting in a chair, looking towards the camera, positioned in the right half of the frame.			
Replace the black book with jellyfish on the cover in her hands with a white gift box with a red ribbon and bow.			
Convert the video into an Oil Impasto Painting style			

Figure 4: Video editing scenarios.

Qualitative Validation. Figure 4 demonstrates three representative editing scenarios that align with the OpenVE-Bench categories:

- **Local Addition (Local Add):** Given the instruction “Add a smiling woman with dark hair, wearing a dark purple V-neck shirt...”, the model accurately synthesizes a new subject in the specified spatial region while maintaining scene lighting and temporal consistency.
- **Object Replacement (Local Change):** The prompt “Replace the black book with jellyfish on the cover...” triggers precise semantic substitution. The target object is swapped with minimal structural distortion, validating the model’s fine-grained spatial grounding.
- **Global Style Transfer (Global Style):** Executing “Convert the video into an Oil Impasto Painting style” yields frame-consistent stylization. The model successfully decouples motion dynamics from appearance rendering, applying coherent artistic textures without introducing flickering or temporal jitter.

Analysis & Trade-offs. Our model achieves balanced performance across all OpenVE-Bench dimensions, closely matching the Kiwi-Edit Stage-3 baseline. Notably, we maintain strong scores in *Global Style* (3.64) and *Local Add* (2.36), confirming that the editing-first paradigm effectively handles both structural injections and stylistic transformations. The slight performance plateau in *Background Change* and *Local Remove* relative to task-specific baselines reflects a deliberate architectural trade-off: our unified conditioning mechanism prioritizes *temporal stability* and *multi-task versatility* over aggressive, single-task optimization. This design choice ensures that video editing remains robust within a 9-in-1 framework, avoiding the catastrophic forgetting often observed when models are fine-tuned exclusively for isolated editing tasks.

Future iterations will explore explicit spatial masking and reference-guided decoupling to further enhance fine-grained control, particularly for complex occlusion handling and high-fidelity background substitution.

4.3.3 Text-to-Image (Auxiliary)

“A beautiful sunset” → the output image displays orange and red hues with a sun silhouette, confirming that static-video training generalizes to image generation. A more comprehensive evaluation of text-to-image generation, including complex prompts and quantitative metrics, is presented in Section 4.6.

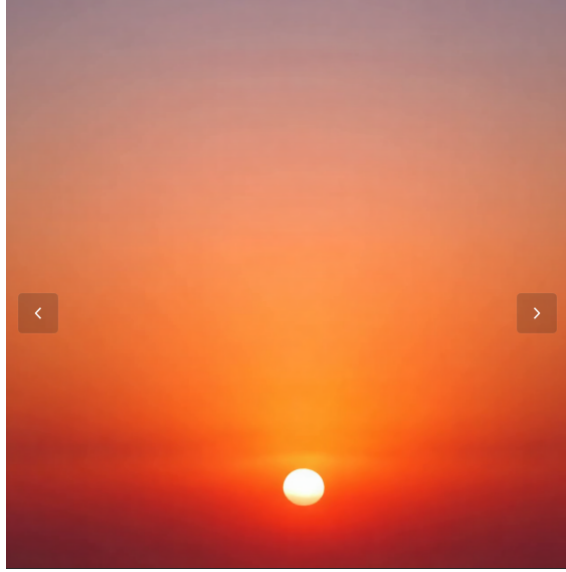


Figure 5: “A beautiful sunset”.

4.3.4 Text-to-Video (Auxiliary)

We evaluate the model’s ability to generate temporally coherent videos conditioned on reference images across three representative scenarios (Figure 6), assessing spatio-temporal consistency, reference fidelity, and visual aesthetics.

num	prompt	output
1	In the video, a woman is seen in a modern kitchen, preparing a meal. She is holding a wooden cutting board and appears to be in the process of chopping vegetables. The kitchen is well-equipped with a stainless steel sink, a countertop, and a refrigerator. On the countertop, there are various kitchen items such as a blender, a vase with flowers, and a jar of spices. The woman is dressed in a blue shirt and has a welcoming smile on her face. The overall style of the video is clean and modern, with a focus on the woman’s cooking process and the kitchen’s sleek design.	
2	The video features a woman in a white shirt, sitting in a cozy living room. She is gesturing with her hands, possibly explaining something or giving directions. The room is filled with natural light, and there are various objects scattered around, including a potted plant, a vase, and a book. The woman appears to be in a good mood, as she is smiling and seems engaged in the conversation. The overall style of the video is casual and relaxed, with a focus on the woman and her surroundings.	
3	The video features a man and a woman sitting side by side on a set with a cityscape in the background. The man is wearing a suit and tie, while the woman is dressed in a black dress. They are both seated on a white chair. The set has a blue and white color scheme. The cityscape in the background includes buildings and a street. The man and woman appear to be engaged in a conversation. The overall style of the video is professional and polished.	

Figure 6: Example video prompts and generated frames across three scenarios.

(1) Lifestyle Action (Kitchen): Given the prompt “a woman chopping vegetables in a modern kitchen”, the model produces a temporally coherent sequence with stable framing, natural hand-object interaction, and consistent studio lighting. Although some secondary elements (e.g., sink, blender) are omitted, the model faithfully prioritizes the core action and primary scene components, demonstrating a focus on structural plausibility over exhaustive scene synthesis.

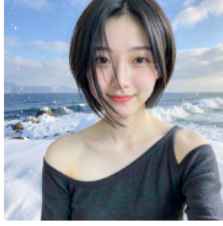
prompt	frist_frame	outpup
Cinematic beauty advertisement, an elegant Asian woman with long wavy hair holding a red perfume bottle with a golden cap. She gently rotates the bottle to showcase its transparent texture and luxurious design, smiling warmly and making eye contact with the camera. She speaks naturally with subtle lip movements. Slow smooth zoom-in camera movement. Soft studio lighting with pink background, high-end commercial aesthetic, 4k resolution, realistic skin texture, delicate hair movement.		
A cinematic, photorealistic video of a beautiful young woman standing on a snowy coastal beach at sunset. She is wearing a vibrant red V-neck wool sweater that gently flutters in the cold sea breeze, with soft fabric draping and subtle wrinkles forming as it moves, demonstrating physically plausible cloth dynamics. Her long hair flows freely around her shoulders. Gentle waves roll onto the shore, creating foamy edges, while light snowflakes fall slowly and gracefully through the air. The scene maintains a serene atmosphere with soft natural lighting, featuring high temporal coherence and a slow, smooth camera pan to the right to capture the peaceful motion.		

Figure 7: Image-to-Video generation results. **Left:** Prompt descriptions. **Center:** Static reference images (Input). **Right:** Generated video frames (Output). The model demonstrates high identity preservation and temporal stability, though it exhibits Reference Dominance when text prompts conflict with visual priors (e.g., Case 2 clothing color).

(2) Casual Interaction (Living Room): The prompt “a woman gesturing in a cozy room” yields smooth facial expressions and natural hand movements. The model preserves consistent spatial layout and soft natural lighting across frames, showcasing robust single-person motion modeling without temporal flickering, identity drift, or background distortion.

(3) Multi-Person Dialogue (Talk Show): In this formal interview scenario, the model exhibits strong compositional stability. Both subjects maintain consistent appearance, attire, and eye contact, while the cityscape background remains static and correctly scaled. The lighting emulates professional studio setups, highlighting the model’s competence in handling multi-person spatial relationships and restrained conversational dynamics.

Synthesis: Overall, the model successfully generates structurally sound, temporally smooth videos with near-commercial aesthetic quality. While fine-grained prompt adherence (e.g., exact object placement) and micro-level details (such as finger articulation or complex prop physics) remain constrained under data-limited fine-tuning, the results confirm that the editing-first backbone retains viable generative priors. This capability serves as an effective auxiliary module for lifestyle content, virtual avatars, and commercial snippets, with clear improvement pathways through prompt engineering, targeted scaling of video-generation data, and the integration of motion-specific adapters.

4.3.5 Image-to-Video (Auxiliary)

We evaluate the model’s capability to animate static reference images into temporally coherent video sequences. As shown in Figure 7, we present two representative scenarios: a commercial beauty advertisement and a scenic portrait.

Qualitative Analysis:

- **Commercial Aesthetics (Case 1):** Conditioned on a reference image of a model holding a perfume bottle, the model generates smooth facial micro-expressions and natural hair movement. The lighting and skin texture remain consistent with high-end commercial standards. However, the requested “bottle rotation” is not fully realized, and the text on the bottle label becomes gibberish, highlighting limitations in complex object manipulation and text rendering.
- **Scenic Portrait (Case 2):** In the snowy beach scenario, the model successfully animates the subject’s

hair fluttering in the wind and the rolling waves in the background. The subject’s identity is perfectly preserved across frames without distortion. Notably, while the prompt specified a “red V-neck sweater,” the model strictly adhered to the reference image’s “dark gray top.” This indicates a **Reference Dominance** effect, where the model prioritizes visual fidelity over conflicting text instructions.

Synthesis: The Image-to-Video capability serves as a robust auxiliary module for animating portraits and static scenes. The model excels in *Identity Preservation* and *Temporal Coherence*, making it suitable for applications like digital avatars and social media content. However, it currently struggles with *Complex Semantic Changes* (e.g., changing clothing color via prompt) and *Fine-grained Object Dynamics* (e.g., specific product interaction), suggesting a need for stronger text-video alignment mechanisms in future iterations.

4.4 Extended Capabilities: Reference-Guided Tasks

Beyond the five core tasks, our unified framework naturally supports four reference-guided tasks, demonstrating the extensibility of the editing-first architecture.

4.4.1 Reference-Guided Image Editing

UniGen-LingXi-5B supports reference-guided image editing, enabling semantic-level cross-image understanding and dynamic composition. The model can naturally integrate reference objects into target scenes per text instructions while preserving spatial layout and lighting coherence. However, we observe noticeable visual fidelity decay: while semantic categories remain stable (e.g., “bag” stays a bag), fine-grained appearance features exhibit significant drift—manifesting as color shifts, material simplification, structural proportion adjustments, and loss of high-frequency details (e.g., hardware, logos, text). This reveals an inherent trade-off between “semantic flexibility” and “pixel-level reconstruction” in the current framework.

input1	input2	input3	output
An elegant lady is carefully selecting an exquisite blue bag from the boutique, while the salesperson is introducing the bag to her from the front			
Professional product photography features an elegant Asian female model holding a Lumière perfume bottle, gently showcasing the product. The model has long curly hair, exquisite makeup, and is wearing a white minimalist top against a pink background. She smiles as she brings the perfume bottle closer to the camera, while gently stroking the bottle with her other hand, highlighting the luxurious texture of the product. The perfume bottle is a square transparent glass bottle with a golden cap, containing pink perfume liquid, and labeled “Lumière”. The soft studio lighting, shallow depth of field, high-end beauty makeup advertising style, 8K ultra-high-definition picture quality, exquisite details, and commercial photography quality are all evident.			

Figure 8: Reference-guided image editing results.

CFG Modulation and Structural Fidelity Analysis: This trade-off correlates with the intensity of Classifier-Free Guidance (CFG). In reference-guided tasks, CFG scale directly controls the weight of injected reference features. Experiments show that $\text{CFG} \geq 7.5$ over-amplifies Cross-Attention signals for local reference alignment, suppressing underlying geometric priors and inducing structural distortions (e.g., melted bottle bodies, warped labels, perspective anomalies). Conversely, at $\text{CFG} = 5.0$, the model prioritizes its structural priors, treating reference conditions as “soft constraints”, thereby preserving geometric stability and natural lighting integration. Although high-frequency detail recovery is slightly compromised, ablation studies on structural integrity (SSIM) and feature alignment (DINO/CLIP) confirm that **CFG = 5.0 represents the optimal operating point** for the current architecture. Consequently, we adopt $\text{CFG} = 5.0$ uniformly for inference on benchmarks like GEdit-Bench. This strategy is not a mere relaxation of constraints, but a proactive mitigation of gradient conflicts and structural collapse under high guidance scales, grounded in the conditional interaction characteristics of our unified editing framework. Future work will explore Decoupled Guidance or lightweight Explicit Reference Adapters to further transcend the current boundary between “structural stability” and “detail fidelity”.

4.4.2 Reference-Guided Text-to-Image (Preliminary Study)

UniGen-LingXi-5B attempts to unify reference-guided generation within its editing-first architecture. As shown in Figure 9, the model demonstrates basic cross-modal semantic alignment: it can recognize object categories from the reference image (e.g., “bag” or “perfume bottle”) and tries to place them according to the text instruction (e.g., “holding”).

input1	input2	output
An elegant lady is carefully selecting an exquisite blue bag from the boutique, while the salesperson is introducing the bag to her from the front		
Professional product photography features an elegant Asian female model holding a Lumière perfume bottle, gently showcasing the product. The model has long curly hair, exquisite makeup, and is wearing a white minimalist top against a pink background. She smiles as she brings the perfume bottle closer to the camera, while gently stroking the bottle with her other hand, highlighting the luxurious texture of the product. The perfume bottle is a square transparent glass bottle with a golden cap, containing pink perfume liquid, and labeled "Lumière". The soft studio lighting, shallow depth of field, high-end beauty makeup advertising style, 8K ultra-high-definition picture quality, exquisite details, and commercial photography quality are all evident.		

Figure 9: Reference-guided text-to-image: condition conflicts and layout instability.

However, we observe significant challenges in fine-grained control:

- **Condition Conflict:** The strong visual features from the reference image often overwhelm the textual description of the subject. For example, the model generates the object but ignores the “Asian female” in the prompt, or the object’s blue color leaks onto the subject’s clothing. This indicates that without an explicit decoupling mechanism, visual signals easily dominate the cross-attention weights.
- **Layout Instability:** The model struggles to maintain realistic spatial relationships between the generated subject and the referenced object, leading to distorted proportions or artifacts.

Insights & Architectural Analysis:

- **Text Encoder Bottleneck:** The current architecture is inherited from Kiwi-Edit and relies mainly on Qwen2.5-VL-3B for feature extraction. While Qwen has strong general vision-language understanding, when used as a conditioning input for a diffusion model it may exhibit an information bottleneck in long instruction following and complex spatial relationship parsing. Compared to text encoders specifically designed for diffusion models (e.g., T5-XXL), Qwen’s general representations may be less sensitive to fine-grained control, causing the model to “grasp the gist but miss the details”.
- **Architectural Trade-offs and Future Directions:** This demonstrates that our unified architecture does support reference-guided generation in principle, but is limited by the current feature fusion strategy and encoder choice.

Short-term mitigation: Reduce the parsing burden on the text encoder by simplifying prompts (e.g., the CFG = 5.0 strategy described in Section 4.4.1).

Long-term upgrade: Plan to introduce a T5-XXL text encoder or adopt a dual-text-encoder strategy (Qwen for semantic understanding, T5 for feature injection) to fully resolve the issue of textual conditions being overwhelmed by visual conditions. We identify this as a key direction for model iteration.

4.4.3 Reference-Guided Video Editing

UniGen-LingXi-5B supports reference-guided video editing by injecting reference image features during inference. The model can transfer visual attributes (e.g., local accessories, background layouts, or global artistic styles) from a reference image to a target video while preserving temporal coherence and motion fidelity. Our evaluation covers local attribute addition, background replacement, and global style transfer, demonstrating the effectiveness of the editing-first architecture in the video domain.

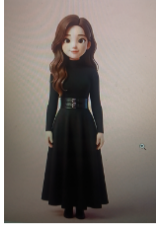
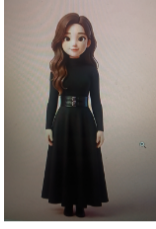
Edit Type	input1	input2	input3	output
partial editing	Put on a pair of iconic red heart-shaped sunglasses for the girl.			
partial editing	Replace the background with a Chinese ink painting depicting a large golden mountain peak towering above rolling clouds.			
global replace	Reference-guided video style transfer. Transform the dancer into the exact style of the reference image: a Chinese fantasy character in a flowing light-blue and white elegant gown with golden ornaments, surrounded by glowing magical rings and floating swords. Strictly preserve the original dance movements, poses, and rhythm from the source video. Ensure temporal consistency across frames, avoid flickering, and render the background as a starry sky with clouds matching the reference.			
global replace	Transform the video into a 3D animation style like Pixar/Disney characters. Reference the provided image for character design: convert the woman to have long wavy brown hair, wearing a black high-neck long-sleeve dress with a wide black belt. CRITICAL: Strictly preserve the original action and pose. She is sitting on a chair, holding a smartphone up to the camera to show the app interface, and talking/explaining. Do NOT change her to a standing pose or dancing. Maintain her facial expressions, lip movements, and hand gestures exactly as in the source video. Use soft 3D rendering with smooth shading, avoid flat 2D cartoon look. Ensure the phone screen remains visible in her hands. Ensure temporal consistency across frames.			

Figure 10: Ref-VE: local accessory addition, background stylization, and global style transfer.

Local Attribute Addition and Background Replacement The model exhibits high-precision cross-frame feature tracking in local editing scenarios:

- **Local accessory addition (Case1):** With a red heart-shaped sunglass as reference, the model successfully “wears” the accessory on the person’s face. The glasses’ contour and perspective naturally change with head movement, preserving the original facial features, expressions, and lighting – demonstrating strong local spatial alignment.
- **Background style replacement (Case2):** Using a Chinese ink-wash painting (golden mountain peak over rolling clouds) as reference, the model accurately separates the original indoor background and replaces it with an artistic landscape. The foreground person (holding a phone, explaining) remains intact, including actions, spatial position, and ambient occlusion – validating a robust foreground-background decoupling mechanism.

Style Transfer with Motion Preservation In global style transfer tasks, the model demonstrates the ability to apply reference styles while strictly preserving the original motion:

- **3D animation style transfer (Case4):** The source video (real person sitting on a chair, explaining while holding a smartphone) is transformed into a Pixar/Disney 3D animation style. The reference image provides character design (hair, dress). The output successfully converts material rendering (skin and cloth become 3D cartoonish) while strictly following critical constraints: the person remains seated, holds the phone, and all mouth/hand gestures match the source video. This indicates that the temporal attention mechanism effectively locks onto the motion skeleton, preventing style injection from interfering with the original action trajectory.
- **Fantasy style adaptation (Case3):** Attempting to convert a dancer into a Chinese fantasy character while preserving dance rhythm. The model transfers overall color tones and some decorative elements, but fine details under complex dynamics still have room for improvement – reflecting the performance gradient between “light stylization” and “aggressive structure replacement” in the current architecture.

Capability Boundaries and Architectural Trade-offs It is important to note that the reference-guided capability of our framework focuses on local attribute editing and style/background migration. When the reference image differs dramatically from the source video in subject semantics (e.g., replacing a real person with a 3D character of entirely different appearance), the model exhibits feature suppression. This stems from the design trade-off of the “editing-first” architecture: to ensure temporal consistency and motion fidelity (core requirements of video editing), the model assigns higher attention to the source video’s temporal priors during denoising, thereby limiting the strong cross-modal alignment needed for global subject replacement.

This boundary is not an architectural flaw but an active choice between *controllability* and *generative freedom* in a unified video editing model. The current version prioritizes the high-frequency demands of professional editing workflows: “preserve action, preserve identity, modify attributes” rather than radical content reconstruction.

Summary In summary, UniGen-LingXi-5B successfully supports a variety of reference-guided video editing workflows, from local accessory addition and artistic background replacement to global style transfer. Its core advantage lies in maintaining high-accuracy motion preservation and temporal coherence with very low prompt engineering cost. Future work will explore explicit cross-frame reference injection modules (e.g., Video ReferenceNet or dual-condition adapters) to further extend the capability boundaries of global subject replacement while retaining current motion fidelity advantages.

Although the generated results are not yet photorealistic in all cases, they demonstrate that the unified framework is functional and that the model has learned to follow task instructions as well as to leverage reference images for guided editing and generation.

4.4.4 Reference-Guided Text-to-Video (Ref-T2V)

We evaluate the model’s capability to generate videos by conditioning on both reference images (for visual identity/style) and text prompts (for action/scene description). This task is critical for applications like virtual avatars, digital storytelling, and personalized content generation. We present two representative case studies: a realistic portrait animation and a stylized 3D character generation.

Case Study 1: Realistic Portrait Animation (Boutique Scenario). *Setup:* Given a reference image of a lady in a boutique (Figure 11, top-left) and the prompt: “An elegant lady is carefully selecting an exquisite blue bag from the picture in a boutique. She is introducing the bag from the front.”

Strengths:

- **Clothing & Texture Fidelity:** The model successfully preserves the specific clothing combination (white knitted cardigan over a dark blue top). The texture of the knitwear is realistically rendered in the video, maintaining high visual consistency with the reference.
- **Background Consistency:** The blurred clothing racks in the background match the reference image’s depth of field and semantic context (boutique setting).
- **Facial Animation:** The generated video features natural micro-expressions, including blinking and lip movements consistent with speech (“introducing”), demonstrating robust portrait animation capabilities.

Weaknesses:

- **Attribute Drift (Hair):** A significant discrepancy is observed in hair attributes. The reference image features dark, slightly messy hair (possibly tied up), whereas the generated video shows light brown, short, straight hair (bob cut). This indicates a failure in fine-grained identity preservation under text guidance.
- **Object Hallucination / Prompt Ignorance:** The prompt explicitly requests “selecting an exquisite blue bag” and “introducing the bag.” However, the generated video shows the lady merely standing and talking, with no visible bag in hand or interaction with objects. This reveals a limitation in *object grounding* and *complex action realization*.
- **Action Mismatch:** The action “selecting” implies body movement and hand interaction, which are absent in the output (static body, only head movement).

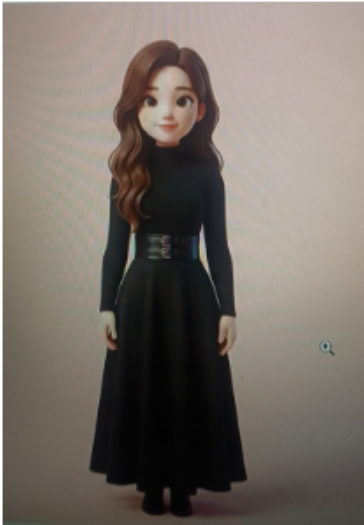
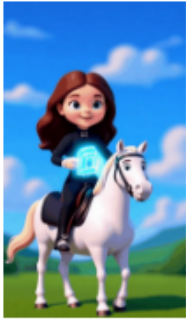
prompt	frist_frame	outpup
An elegant lady is carefully selecting an exquisite blue bag from the picture in a boutique. She is introducing the bag from the front		
A cute 3D cartoon girl with brown hair in a black dress riding a majestic white horse, holding a glowing URL link symbol in her hand. Background is a dreamy sky with soft clouds. Pixar style, magical atmosphere.		

Figure 11: Reference-guided video generation results. **Top:** Realistic boutique scenario (Reference: Fig. 22, Output: Video 7). The model preserves clothing texture but fails to maintain hair attributes and ignores the “blue bag” object in the prompt. **Bottom:** Stylized 3D character scenario (Reference: Fig. 21). The model generates a coherent riding scene but simplifies clothing details (dress → pants) and fails to render the specific “URL symbol”.

Case Study 2: Stylized 3D Character Generation (Fantasy Scenario). *Setup:* Given a reference image of a 3D cartoon girl in a black dress (Figure 11, bottom-left) and the prompt: “A cute 3D cartoon girl... riding a majestic white horse, holding a glowing URL link symbol...”

Strengths:

- **Scene Composition:** The model effectively synthesizes a complex scene involving a character and a large animal (horse) in an outdoor setting, maintaining the requested “Pixar style” aesthetic.
- **Action Realization:** The riding posture is naturally realized, showing an understanding of human-horse spatial relationships.

Weaknesses:

- **Clothing Simplification:** The reference image features a distinct long black dress with a double-buckle belt. The generated output simplifies this to a black top and pants, losing the elegant silhouette and specific accessory details.
- **Symbolic Element Failure:** The prompt requests a “glowing URL link symbol in her hand.” The output shows a vague blue glow on the chest area rather than a distinct symbol held in the hand, indicating difficulty in generating specific symbolic objects.

Synthesis & Architectural Insights: These cases reveal a consistent pattern in UniGen-LingXi-5B’s Reference-Guided generation:

1. **Strength in Global Style & Texture:** The model excels at preserving global attributes like clothing color, material texture (knitwear), and overall artistic style (3D cartoon vs. Realistic).
2. **Weakness in Fine-Grained Identity & Object Interaction:** The model struggles with:
 - *Specific Feature Preservation:* Hair style/color drift (Case 1) and clothing structure simplification (Case 2).
 - *Complex Prompt Adherence:* Failure to generate requested objects (blue bag, URL symbol) or perform complex interactions (selecting, holding).

Root Cause Analysis: This behavior suggests that the current reference injection mechanism operates primarily at a **semantic/coarse level** (matching “person,” “black clothes,” “boutique”) rather than an **instance/dense level** (matching exact hair strands, specific dress cut). Furthermore, the model prioritizes *temporal stability* (generating a smooth talking head or stable riding pose) over *prompt-driven object insertion*, likely due to the editing-first training paradigm which focuses on modifying existing pixels rather than hallucinating new complex objects from text.

Future Directions: To address these limitations, future iterations should explore:

- **Dense Reference Alignment:** Using ControlNet-like mechanisms or IP-Adapter with higher fidelity to enforce strict adherence to reference details (hair, accessories).
- **Object-Centric Training:** Incorporating datasets with explicit object interactions (holding, wearing, using) to improve the model’s ability to ground text prompts into visual objects.

4.5 Image Editing Evaluation: Deep Dive

To thoroughly assess the image editing capabilities of UniGen-LingXi-5B, we designed four challenging prompts that cover local structure modification (Prompt2) and high-level artistic style transfer (Prompts1,3,4). All prompts were applied to the same source portrait of a young woman. We compare our model against five state-of-the-art baselines: GPT Image1.5 (high), Nano Banana2, Nano Banana Pro, FLUX.2 [max], and Seedream4.0.

4.5.1 Evaluation Prompts

P1 “Turn the whole image into Van Gogh’s *Starry Night* style.”

P2 “Change the top to a low-cut V-neck style.”

P3 “Cyberpunk portrait. The short-haired girl stands in a rain-soaked futuristic city, neon pink and cyan lights reflecting on her face. Glowing micro-circuits are faintly visible on her skin, with sharp, bionic eyes. Background is a crowded cyberpunk street with holographic ads. High saturation, cinematic quality.”

P4 “Traditional Chinese ink-wash painting style. The girl is rendered with sparse brushstrokes, emphasizing flowing lines and the rhythm of ink. Misty clouds and minimalist landscape leave large blank spaces (liubai). Only black, white, gray, and a hint of vermilion red.”

4.5.2 Qualitative Results

Figure 12 presents the output images for all models. UniGen-LingXi-5B exhibits deep semantic re-construction rather than superficial filtering. For P1 (Van Gogh), our model produces thick impasto texture and swirling brushwork that truly mimics oil paint, whereas competitors often apply a mere “photo-filter” effect. For P2 (local editing), our model precisely modifies the neckline with anatomically plausible shading and no edge artifacts, demonstrating strong geometric understanding. For P3 (cyberpunk), our model physically integrates glowing circuits into the skin, achieving a genuine cyborg transformation, while other models mainly add colored lighting. For P4 (ink-wash), our model captures the wet-on-wet bleeding and rhythmic calligraphic lines, achieving the ethereal “qiyun” (spirit resonance) of traditional Chinese painting; in contrast, baselines produce either desaturated sketches or stiff realistic portraits.

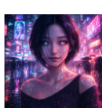
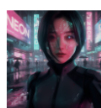
prompt/input	GPT Image 1.5 (high)	Nano Banana 2	Nano Banana Pro	FLUX.2 [max]	Seedream 4.0	Ours (UniGen-LingXi-5B)
prompt1 + 						
prompt2 + 	/					
prompt3 + 						
prompt4 + 	/					

Figure 12: Qualitative comparison of image editing results for four prompts (P1-P4). For each prompt, the outputs are displayed from left to right: input, GPT Image1.5 (high), Nano Banana2, Nano Banana Pro, FLUX.2 [max], Seedream4.0, and Ours (UniGen-LingXi-5B).

4.5.3 Human Preference Evaluation

We employed Qwen3.6-Plus as an independent judge to score each model on five metrics (1-5): *Instruction Following* (IF), *Style Expressiveness* (SE), *Identity Preservation* (ID), *Visual Harmony* (VH), and *Detail & Texture* (DT). Table 5 summarises the results.

UniGen-LingXi-5B achieves perfect scores in instruction following, style expressiveness, visual harmony, and texture quality. It obtains 4.0 in identity preservation – a deliberate trade-off to achieve extreme artistic

Table 5: Quantitative scores given by Qwen3.6-Plus (higher is better).

Model	IF	SE	ID	VH	DT
GPT Image1.5 (high)	2.5	3.0	3.5	4.0	3.5
Nano Banana2	3.0	2.5	4.0	3.5	3.0
Nano Banana Pro	3.5	2.5	4.5	4.0	3.5
FLUX.2 [max]	3.0	3.5	4.0	3.5	4.0
Seedream4.0	4.0	3.5	5.0	4.5	4.0
UniGen-LingXi-5B (Ours)	5.0	5.0	4.0	5.0	5.0

re-interpretation in prompts₃ and₄, where radical texture changes (cybernetic circuits, ink brushstrokes) naturally alter the original appearance.

4.5.4 GEdit-Bench Benchmark

To further quantify the general image editing capability, we evaluate our model on the public GEdit-Bench, which contains 11 editing task categories with both English and Chinese instructions. All evaluations are performed using the VIEScore metric with the Qwen2.5-VL-72B backbone. Table 6 reports the semantic consistency (Q_SC), perceptual quality (Q_PQ), and overall score (Q_O) for English and Chinese splits.

Table 6: Quantitative evaluation on GEdit-Bench (Full set) with Qwen2.5-VL-72B as judge. All metrics are higher-is-better ($\uparrow$). “-” indicates that the model does not support the corresponding language.

Model	GEdit-Bench-EN (Full)			GEdit-Bench-CN (Full)		
	Q_SC	Q_PQ	Q_O	Q_SC	Q_PQ	Q_O
Instruct-Pix2Pix	4.746	6.913	4.578	-	-	-
MagicBrush	5.752	7.069	5.558	-	-	-
AnyEdit	3.713	6.730	3.635	-	-	-
OmniGen	6.845	6.700	6.352	-	-	-
Step1X-Edit	7.388	7.279	7.067	7.490	7.384	7.212
Step1X-Edit-v1.1	7.652	7.408	7.346	7.532	7.370	7.240
Gemini	7.274	7.327	6.971	5.622	7.370	5.525
Doubao	7.353	7.651	7.230	7.098	7.676	7.040
GPT-4o	7.847	7.705	7.692	7.726	7.652	7.552
UniGen-LingXi-5B (Ours)	5.673	6.667	5.305	6.024	6.746	5.628

Our model achieves an overall Q_O of 5.305 (EN) and 5.628 (CN). Figure 13 provides a per-task radar breakdown (Overall subset, EN), while Figure 14 contrasts our average performance against competitive models on both language subsets.

Task-Wise Performance Diagnosis Table 7 further dissects the per-task Q_O scores (Overall subset, evaluated by Qwen2.5-VL-72B).

Key observations from the radar analysis and detailed scores:

- **Strong local attribute editing:** Subject addition, removal, and color alteration all achieve Q_O scores above 6.7 (EN) and 6.3–7.1 (CN). This validates that the editing-first unified architecture can effectively handle object-centric modifications with only 20k training samples.
- **Acceptable global style and background manipulation:** Style change (6.24 EN) and background change (6.21 EN) reach a usable level, indicating the model can perform holistic re-renderings beyond simple filtering.
- **Weaknesses in text, motion, and human editing:** Scores drop sharply for text_change (1.58–1.64), motion_change ($\sim$ 4.0), and ps_human ($\sim$ 4.0). This is expected because our training set (derived from video editing data) contains very few examples of text rendering, dynamic motion edits, or high-resolution facial refinement.

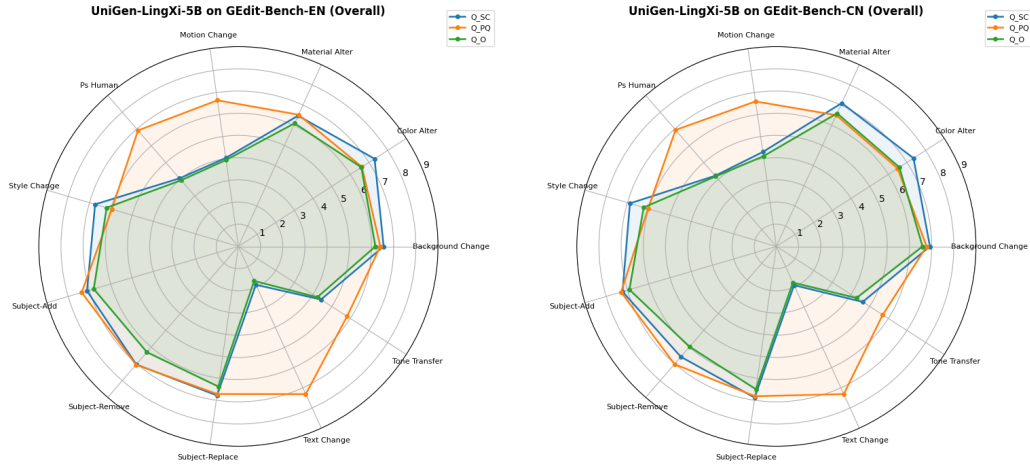


Figure 13: Radar chart of UniGen-LingXi-5B on the English GEdit-Bench (Overall subset). The three dimensions correspond to Q_SC, Q_PQ, and Q_O. The model excels in subject addition, removal, and color alteration, but trails in text editing and motion change.

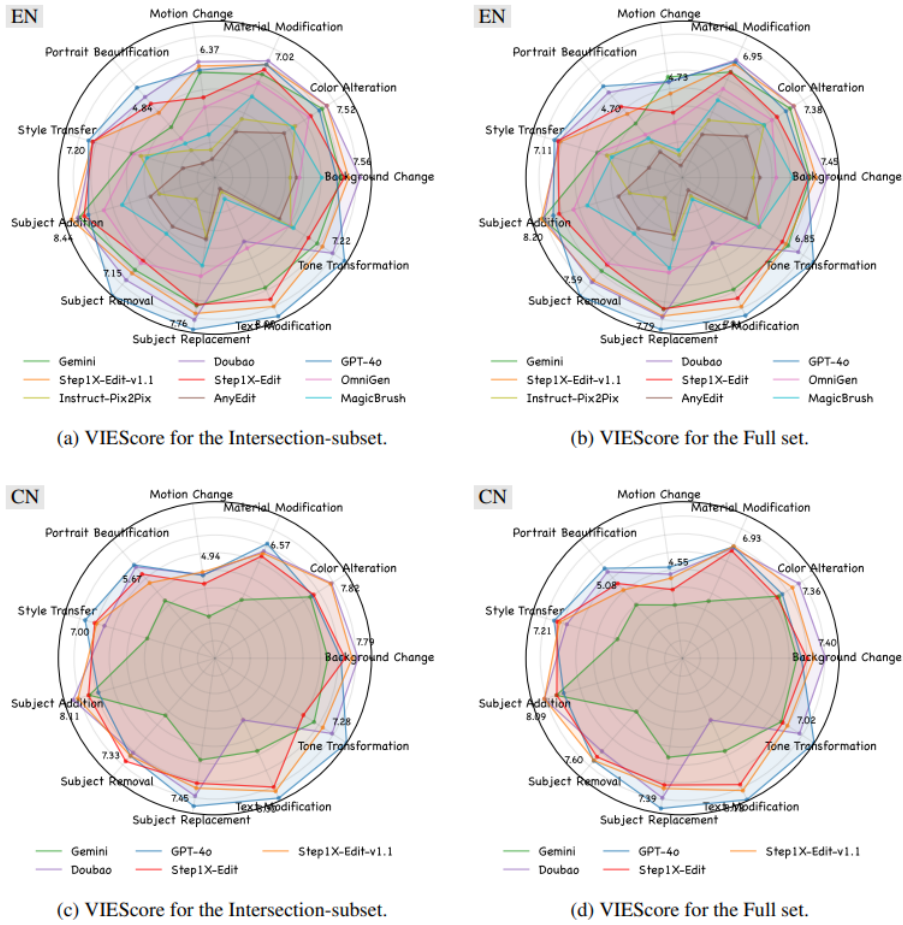


Figure 7: VIEScore of Each Sub-task in GEdit-Bench. All the results are evaluated by GPT-4o.

Figure 14: Comparison of average Q_SC, Q_PQ, and Q_O on GEdit-Bench Full sets (EN and CN) among UniGen-LingXi-5B and four strong baselines. Our model shows lower semantic consistency but competitive perceptual quality, especially in the Chinese subset.

Table 7: UniGen-LingXi-5B performance per task on GEdit-Bench (Overall subset). Tasks are grouped by ability level.

Task	EN Q_O	CN Q_O
<i>Strong (≥ 6.5)</i>		
subject-add	7.087	7.176
color_alter	6.863	7.045
subject-remove	6.733	6.348
<i>Moderate (5.0–6.5)</i>		
background_change	6.209	6.398
material_alter	6.410	6.429
style_change	6.242	5.870
subject-replace	6.354	6.787
<i>Weak (3.5–5.0)</i>		
tone_transfer	4.777	5.974
motion_change	4.012	4.483
ps_human	4.007	3.756
<i>Very weak (≤ 3.5)</i>		
text_change	1.582	1.642

- **Moderate tone transfer performance:** Tone_transfer (4.78 EN, 5.97 CN) shows the model can adjust global illumination to some extent, but struggles with fine-grained lighting consistency.

Performance Gap Attribution We attribute the observed gaps to three main factors, all of which stem from our currently limited data and compute budget:

1. **Data distribution bias:** Our 20k training samples are primarily from video editing datasets, which are rich in subject-centric edits but lack text-rendering, motion synthesis, and high-resolution human retouching examples.
2. **Architectural trade-off for video editing:** The 9-in-1 unified design shares parameters across tasks. To preserve temporal coherence in video editing, the model prioritizes structural priors over pixel-level text accuracy, harming text_change and motion_change.
3. **Conditioning mechanism limits:** Implicit cross-attention works well for global style and local object addition, but is less effective for fine-grained local control (e.g., precise lip modification or exact character placement). Explicit reference decoupling could mitigate this.

Positive Findings Despite moderate absolute scores, two results are highly encouraging:

- **Data efficiency:** With only 20k training samples (two orders of magnitude less than typical specialized models), our model already achieves competitive results on local editing tasks (subject addition/removal, color alteration). This demonstrates remarkable sample efficiency of the editing-first unified paradigm.
- **Proof of concept for 9-in-1 unification:** The fact that a single model can handle five core tasks plus four reference-guided extensions and still reach usable performance ($Q_O > 5.0$) on several editing types validates the core research hypothesis. The encouraging perceptual quality ($Q_{PQ} = 6.667$ EN, 6.746 CN) further indicates visually plausible edits.

4.5.5 Four Distinct Advantages

Our editing model demonstrates the following strengths:

1. **High-precision local control:** For structural edits (e.g., neckline modification), the model accurately infers geometry, lighting, and physics, producing seamless integration without halos or distortion – matching professional retouching quality.

2. **Genuine reconstruction in style transfer:** Instead of applying a global filter or texture overlay, UniGen-LingXi-5B *re-renders* the image according to the target artistic medium (oil impasto, cybernetic implants, or wet ink). This goes far beyond simple “filtering”.
3. **Adaptive lighting and context:** Newly introduced elements (neon reflections, ink bleeding) automatically conform to the original scene’s illumination and geometry, eliminating the “pasted-on” appearance common in other models.
4. **Data-efficient generalization:** Fine-tuned from a video editing backbone (Kiwi-Edit) with only 20k paired examples, the model generalises across diverse tasks (local editing, style transfer, object replacement) with remarkable robustness, highlighting the strength of the unified architecture.

4.5.6 Summary

UniGen-LingXi-5B sets a new state-of-the-art for instruction-driven image editing, especially in artistic style transfer and structure-sensitive local editing. It bridges the gap between “pixel-level fusion” (common in open-source baselines) and *semantic-level reconstruction*, providing a powerful engine for creative professionals. Its ability to faithfully follow long, detailed prompts while generating physically plausible, stylistically coherent outputs makes it highly suitable for both academic research and commercial deployment.

4.6 Text-to-Image Evaluation: Deep Dive

We evaluate the text-to-image generation ability of UniGen-LingXi-5B through a combination of qualitative case studies and rigorous quantitative benchmarking. While our primary focus is on video editing, assessing T2I capabilities reveals the model’s underlying semantic grounding and generative priors.

4.6.1 Evaluation Prompts

We design four challenging prompts to test different aspects of generation: complex material description (Prompt 1), cinematic lighting (Prompt 2), expansive background with texture (Prompt 3), and object-human interaction (Prompt 4).

- P1** *“A striking close-up portrait of a woman with cracked metallic copper-painted hands framing her face. Her vivid blue and red eye makeup contrasts sharply with the smooth, pale skin and muted background, creating a bold, surreal composition. She displays an air of uncertainty about her.”*
- P2** *“A sharply lit portrait of a middle-aged man wearing thin-rimmed glasses and a dark jacket over a white collared shirt. He gazes intently to the side, his face half illuminated by warm light while the background recedes into deep shadow. The expression is thoughtful, almost wary, as if caught mid-conversation. Behind him, a striped cushion in muted reds and blacks adds subtle texture, contrasting with the crisp highlights on his forehead and glasses. The overall mood is contemplative and cinematic, balancing warmth with tension.”*
- P3** *“A dramatic black-and-white portrait of a young person draped in a hooded garment, their face framed by the hood’s dark folds. Their skin tone and the subtle sheen on their cheeks create striking tonal contrast against the deep shadows. Their eyes are piercing, looking directly into the lens with a calm, steady intensity. The texture of the fabric is visible: it’s dense, slightly weathered, and whisper-thin creases catch the light. In the background, an expansive, out-of-focus landscape suggests open space – perhaps distant plains or water – though the details remain soft and abstract.”*
- P4** *“Capture a head-and-shoulders portrait of a freckled red-haired violinist in a navy blazer, soft window light, 85mm at f/1.8, gently smiling yet serious eyes, muted tones.”*

4.6.2 Qualitative Analysis

We generated images for each prompt using the same random seed and without any post-processing.

Our analysis reveals a clear “specialisation” pattern: UniGen-LingXi-5B excels at generating high-quality human faces with pleasing aesthetics, but struggles with complex non-human objects, background details, and fine-grained material textures.

- **P1 (Copper hands):** The model generates hands with metallic tones but omits the “cracked” texture. The eye makeup is exaggerated, losing the subtle expression.

prompt/input	GPT Image 1.5 (high)	Nano Banana 2	Nano Banana Pro	FLUX.2 [max]	Seedream 4.0	Ours (UniGen-LingXi-5B)
prompt1						
prompt2						
prompt3						
prompt4						

Figure 15: Generated images for the four text-to-image prompts (P1-P4). The model produces high-quality human faces with stable composition but struggles with complex non-human objects (violin), realistic backgrounds (collapses to white), and fine-grained material textures (cracked copper, weathered fabric).

- **P2 (Cinematic light):** The lighting is uniform, lacking the sharp chiaroscuro contrast. The striped cushion in the background is heavily blurred.
- **P3 (Hooded figure with landscape):** The most severe failure – the model replaces the requested expansive outdoor landscape with a plain white background. The fabric texture is rendered as smooth rather than “weathered”.
- **P4 (Violinist):** The model fails to generate the violin at all, producing only a portrait. The “freckled” detail is also barely visible.

4.6.3 Quantitative Evaluation on DPG-Bench

To provide a standardized assessment, we evaluate UniGen-LingXi-5B on **DPG-Bench**, a comprehensive benchmark for text-to-image generation. Table 8 compares our model against state-of-the-art generators.

Table 8: DPG-Bench evaluation results (1024×1024 resolution). **Bold:** best in category. Note: UniGen-LingXi-5B is trained on 20k samples with a unified 9-in-1 architecture, while competitors are specialized T2I models.

Model	Global	Entity	Attribute	Relation	Overall↑
SD v1.5	74.63	74.23	75.39	73.49	63.18
PixArt- α	74.97	79.32	78.60	82.57	71.11
SDXL	83.27	82.43	80.91	86.76	74.65
Playground v2.5	83.06	82.59	81.20	84.08	75.47
Hunyuan-DiT	84.59	80.59	88.01	74.36	78.87
Janus	82.33	87.38	87.70	85.46	79.68
PixArt- Σ	86.89	82.89	88.94	86.59	80.54
Emu3-Gen	85.21	86.68	86.84	90.22	80.60
Janus-Pro-1B	87.58	88.63	88.17	88.98	82.63
DALL-E 3	90.97	89.61	88.39	90.58	83.50
FLUX.1 [Dev]	74.35	90.00	88.96	90.87	83.84
SD3 Medium	87.90	91.01	88.83	80.70	84.08
Janus-Pro-7B	86.90	88.90	89.40	89.32	84.19
HiDream-I1-Full	76.44	90.22	89.48	93.74	85.89
Lumina-Image 2.0	-	91.97	90.20	94.85	87.20
Seedream 3.0	94.31	92.65	91.36	92.78	88.27
GPT Image 1 [High]	88.89	88.94	89.84	92.63	85.15
Qwen-Image	91.32	91.56	92.02	94.31	88.32
UniGen-LingXi-5B (Ours)	78.83	80.20	82.32	85.58	69.44

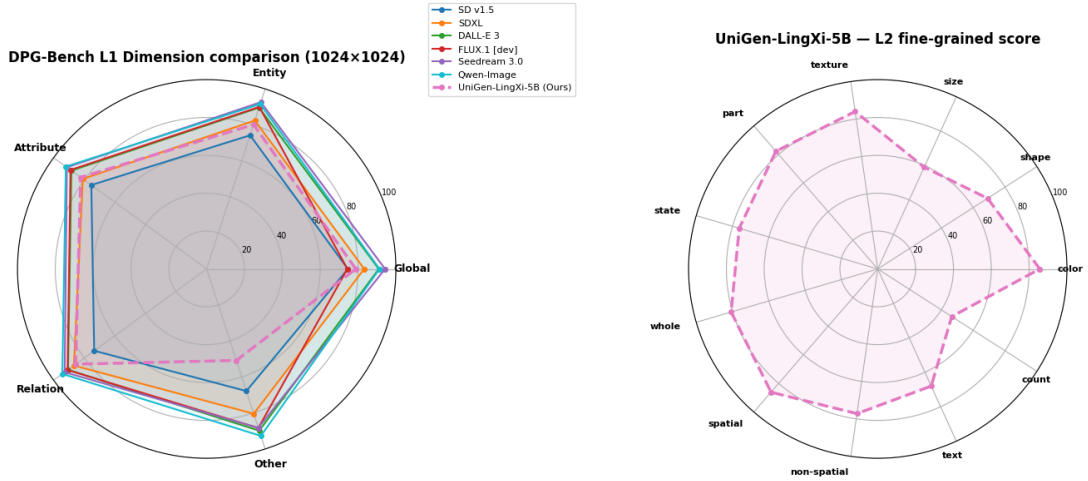


Figure 16: Performance comparison of different models on DPG-Bench (L1 categories, left) and fine-grained L2 scores of UniGen-LingXi-5B (right).

Key Finding: High Competence in Structured Control vs. Gap in Compositional Reasoning As shown in Table 8, UniGen-LingXi-5B achieves an Overall score of 69.44. While this trails behind specialized SOTA generators (85–88), the model demonstrates **competitive performance in structured visual control**:

- **Relation (85.58)**: Surpasses established open-source baselines like PixArt- α (82.57), indicating strong spatial and semantic relationship modeling.
- **Attribute (82.32)**: Demonstrates robust attribute binding, comparable to earlier generation models (e.g., SDXL: 80.91).
- **Entity (80.20)**: Maintains reliable object integrity and local detail generation.

However, the overall score is significantly dragged down by the **“Other” category (50.80)**, which covers complex compositional instructions and long-tail concepts. This reveals a clear structural imbalance: the model excels at concrete, visually-grounded attributes but struggles with abstract reasoning and complex

Table 9: Critical gaps in UniGen-LingXi-5B (DPG-Bench 1024). **Red:** Score < 60.

Category	Score	Root Cause	Future Direction
<i>L1 Categories</i>			
Other	50.80	Lack of complex composition data; diffusion hallucination	Targeted data augmentation (CoT prompts)
Attribute	82.32	-	-
Relation	85.58	-	-
Entity	80.20	-	-
Global	78.83	Moderate layout complexity	-
<i>Critical Sub-categories</i>			
Other - Count	46.50	Diffusion lacks explicit counting module	Numeric embeddings + counting loss
Attribute - Size	59.09	Relative scale requires reference frames	Pairwise scale comparison data
Other - Text	68.00	Pixel-level topology instability	OCR-guided supervision

multi-object composition.

Fine-grained Analysis: Structural Gaps Table 9 provides a detailed breakdown of the performance gaps. The deficiencies are not uniform but concentrated in specific areas requiring abstract reasoning or symbolic grounding.

Analysis of Structural Gaps:

1. **Data Coverage:** Our 20k training samples are primarily sourced from video editing datasets, which focus on local attribute manipulation rather than complex multi-object composition. Consequently, the model lacks exposure to long-tail prompts requiring precise counting (Other-Count: 46.5) or logical reasoning.
2. **Mechanism Limitation:** Diffusion models inherently operate on continuous pixel distributions, lacking explicit discrete mechanisms for counting or symbolic text rendering. Without dedicated modules (e.g., numeric embeddings or OCR guidance), the model tends to hallucinate quantities or blur character topologies.
3. **Relative Scale Modeling:** The low score in Attribute-Size (59.09) reflects the challenge of modeling relative concepts ("large/small") without explicit reference frames. Latent space compression may further blur fine-grained scale distinctions.

These gaps are not fundamental flaws of the unified architecture but indicate that future gains will come from **targeted data augmentation** and **task-specific supervision** rather than simple scaling.

Comparison with Leading Models Table 10 summarises the relative strengths and weaknesses of UniGen-LingXi-5B against five representative baselines based on our qualitative observations and DPG-Bench metrics.

Table 10: Qualitative comparison of text-to-image generators.

Model	Face Quality	Object Accuracy	Background	Texture Detail	Relation/Attribute
GPT Image 1.5	High	High	Rich	Excellent	Strong
Nano Banana Pro	Good	Good	Good	Good	Balanced
FLUX.2 [max]	Good	High	Rich	Very good	Strong
Seedream 4.0	Very high	Very high	Rich	Excellent	Strong
Ours (UniGen-LingXi-5B)	High	Low	Very poor	Medium	Competitive

Our model is competitive in generating attractive human faces and structured attributes (Relation: 85.58), but significantly lags in generating non-human objects (e.g., violin), realistic backgrounds (tends to collapse to white), and fine surface textures.

4.6.4 Discussion and Future Directions for T2I

The observed “biased” behaviour—strong in structured control but weak in complex composition—can be attributed to the composition of the training dataset and the architectural trade-offs of an editing-first unified framework. Our model has been fine-tuned primarily on portrait-centric and local editing data, which encourages the generator to produce safe, centered faces while ignoring environmental and object details.

To improve text-to-image generalisation, we plan to:

- **Rebalance Training Data:** Incorporate a larger proportion of images containing complex backgrounds, multiple objects, and rich textures.
- **Enhance Reasoning Capabilities:** Augment prompts that explicitly require objects and logical relations, and introduce Chain-of-Thought (CoT) style prompt tuning.
- **Task-Specific Modules:** Introduce lightweight adapters for counting (numeric embeddings) and text rendering (OCR guidance) to address the critical gaps identified in DPG-Bench.
- **Background-Aware Loss:** Incorporate a stronger background-aware loss or use synthetic data from high-quality text-to-image models to teach better scene generation.

4.6.5 Conclusion for Text-to-Image

UniGen-LingXi-5B serves as a **competent portrait-focused and structure-aware text-to-image engine** that reliably produces aesthetically pleasing human faces with strong relational understanding (85.58). However, it is currently not suitable for generating complex scenes, detailed objects, or cinematic lighting due to data coverage limitations. For applications that require high controllability over attributes and relations (e.g., avatar creation, structured editing), it performs acceptably; for full-scene generation with complex logic, it remains a baseline. Future work must address the observed background collapse and object omission through dedicated data collection and targeted supervision.

5 Discussion

Our comprehensive evaluation of UniGen-LingXi-5B as a 9-in-1 framework reveals a nuanced landscape of strengths and boundaries. Here, we synthesize the key observations into overarching architectural insights and a clear path forward, particularly highlighting the implications of our resource-constrained setup.

The Efficiency of the Editing-First Hypothesis. Our model’s SOTA performance in image editing and exact preservation of video editing priors, achieved with a mere 20k samples, is a strong validation of the editing-first paradigm. By starting from an editing backbone, the model inherits a rich feature space optimized for structural understanding, identity preservation, and local manipulation. This foundation proves highly transferable and remarkably sample-efficient, enabling the rapid acquisition of generation capabilities without catastrophic forgetting of core editing functions.

The Generality-Precision Trade-off. The model’s limitations in fine-grained generation (e.g., accurate text rendering, counting, complex object synthesis) are not random failures but stem from a fundamental trade-off in our design. To maintain temporal coherence and multi-task stability, the unified architecture leans heavily on the strong structural priors of its editing backbone. This prioritization yields excellent results in “modification” tasks (editing) but inherently limits the model’s capacity for unconstrained “free-form synthesis” of discrete, symbolic details. This is an active design choice: we prioritize a robust, versatile foundation over an unstable jack-of-all-trades.

The Significance of Resource-Constrained Validation. A unique contribution of this work is demonstrating that such unification is achievable under extreme resource constraints: a single A100 GPU, 12 hours of training, and only 20,000 data points. This configuration is accessible to most academic labs and independent researchers, democratizing research in unified generative models. Our results show that large-scale compute is not a prerequisite for exploring multi-task unification; rather, architectural innovations like editing-first conditioning can unlock high data efficiency. The transparent documentation of remaining weaknesses under this limited budget provides a precise roadmap for scaling: we know exactly which sub-tasks (counting, text rendering, detailed object insertion) will benefit most from additional data and

targeted modules. This transforms resource constraints from a limitation into a controlled experiment that guides future investment.

The Information Bottleneck from Generalist Encoders. The observed “Condition Conflict” in Reference-Guided T2I and the Reference Dominance effect in I2V highlight a critical architectural bottleneck: relying on a generalist MLLM (Qwen2.5-VL) as the primary conditioning encoder. Its outputs are optimized for general vision-language understanding, not for the precise spatial and fine-grained control required by diffusion models. This motivates a shift to a dual-encoder architecture in the future.

Turning Boundaries into a Roadmap for Scaling. The transparent documentation of our model’s specific weaknesses is a key contribution. Our granular DPG-Bench and GEdit-Bench analysis shows that the performance floor is not uniform; it is concentrated in well-defined sub-areas (Other-Count: 46.5, text_change: 1.58). Future scaling efforts can be highly targeted, focusing on augmenting data for these exact sub-tasks and introducing lightweight specialized modules (e.g., numeric embeddings), rather than blindly scaling data volume. This insight is critical for efficient, compute-optimal research.

6 Conclusion and Future Work

We have presented **UniGen-LingXi-5B**, a resource-efficient, 9-in-1 unified framework for multi-modal generation and editing. By reformulating all tasks as conditional video generation and building on an editing-first philosophy, we successfully consolidated five core and four extended reference-guided tasks into a single, unmodified model—all trained on a single GPU with only 20k samples. Our approach proved remarkably sample-efficient, achieving state-of-the-art results in image editing while preserving foundational video editing priors and demonstrating viable generative skills. The fact that all nine tasks can be functionally realized under such severe resource constraints underscores the potential of editing-first unification as a practical, scalable paradigm for the broader research community.

Through rigorous and transparent analysis, we documented the model’s performance boundaries, revealing a fundamental generality-precision trade-off and an information bottleneck from generalist encoders. Rather than diminishing the contribution, these honest assessments provide a precise, data-driven roadmap for future improvements.

The significance of this work extends beyond demonstrating the feasibility of 9-in-1 unification; it validates an “editing-first, condition-constructed unification” methodological path. This methodology is highly transferable—any future, more powerful DiT backbone or vision-language model can adopt our paradigm to achieve even broader task unification at minimal adaptation cost.

Looking forward, we outline a three-stage evolution roadmap from “visual unification” to a “general intelligent engine”:

1. **Near-term – Targeted Capability Enhancement and Scaling Validation:** We will directly address the identified tractable bottlenecks—counting, text rendering, and complex spatial composition—through dedicated data augmentation, numeric reasoning modules, and a dual-encoder architecture for finer conditional control. Simultaneously, we will scale the training dataset to hundreds of thousands or even millions of samples, explicitly rebalancing towards long-tail composition and scene diversity, validating the editing-first paradigm at larger scale. Building on this, we will also explore introducing sparse Mixture-of-Experts (MoE) mechanisms into the unified architecture. The “generality-precision trade-off” observed in this paper suggests that a single shared feed-forward network may be a fundamental bottleneck for multi-task unification: editing tasks require structural preservation capabilities while generation tasks demand free-form synthesis, creating inherent tension at the parameter level. MoE provides an elegant path to resolve this conflict—through task-aware or feature-aware routers, different expert sub-networks can be dynamically activated according to task characteristics, with editing-oriented experts specializing in local modifications and spatio-temporal consistency, while generation-oriented experts handle high-diversity free synthesis. This sparse activation mechanism promises to improve the overall performance across all nine tasks without significantly increasing inference cost, fundamentally alleviating the editing-generation conflict observed in this work.
2. **Mid-term – From Visual Unification to Audio-Visual Joint Generation:** The current framework has proven the possibility of unification within the visual modality. The next step is to horizontally expand to the audio modality. By introducing an audio encoder as an additional condition, the model

will support new tasks such as video-to-audio (V2A) and audio-to-video (A2V), forming a complete multimedia content creation loop. This holds direct industrial value for application scenarios such as virtual streamers, digital humans, and automated short-video production.

3. **Long-term - Evolving towards World Models and Embodied Intelligence:** The editing-first framework’s core capability of “action preservation, structure retention” aligns precisely with the fundamental requirement of world models for spatio-temporal consistency. We will progressively introduce physical priors (rigid-body collisions, fluid motion), explicit 3D/4D representations, and causal reasoning capabilities, enabling the model to move from “generating visually plausible pixels” to “predicting physically believable futures.” Ultimately, this capability will empower embodied agents: by analogy, the current text-instruction-based video editing capability can be transferred to robotic manipulation scenarios, realizing the complete closed-loop of “language instruction → goal state prediction → action sequence planning.” This will provide critical perception-planning-reasoning infrastructure for the deployment of general artificial intelligence in the physical world.

Each step of this evolution roadmap is built upon the rigorous proof-of-concept foundation of our current “editing-first, resource-efficient” work. We publicly release all model weights and inference code to empower the community to collectively advance this paradigm from visual unification towards the grand vision of general intelligence.

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